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TCP and UDP Ports used by FactoryTalk Optix

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Summary

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Question

What are all the TCP/UDP ports used by FactoryTalk Optix?

Environment

FactoryTalk Optix

Answer

The following ports are required to properly configure firewalls and port forwarding:

Native modules:

- WebPresentationEngine (user-configurable)
 - If the protocol is set to HTTP -> 80/TCP
 - If the protocol is set to HTTPS -> 443/TCP
- ApplicationUpdateServer (fixed) -> 49100/TCP
- Communication Drivers:
 - RA EtherNet/IP Driver -> 44818/TCP
 - Modbus -> 502/TCP

- Melsec FX3U -> TCP port needs to be configured on PLC (1034/TCP default in Optix)
- Melsec Q -> TCP port needs to be configured on PLC (1/TCP default in Optix)
- Omron EtherNet/IP -> 44818/TCP
- Omron FINS -> 9600/TCP
- S7 TCP -> 102/TCP
- S7 TIA -> 102/TCP
- TwinCAT -> 851/TCP
- CODESYS -> 1217/TCP
- DataStores (user-configurable)
 - SQL/SQL Express -> 1433
 - MySQL -> 3306
- User Authentication
 - LDAP -> 389 (default, configurable in Domain server address)
 - AD -> 389 (if necessary)
- OPC/UA
 - Server (user-configurable) -> 59100/TCP
 - Client (user-configurable) -> Depends on source server configuration
- FactoryTalk Remote Access Service
 - 5000 / 14273 TCP
- FactoryTalk Hub
 - FactoryTalk Optix acts as a client, a random outbound port is used

External modules (from TemplateLibrary):

- MQTT (PushAgent and MQTTLogic):
 - Standard -> 1883/TCP
 - Secure -> 8883/TCP
- Email (AlarmsWithEmailNotification):
 - SMTP -> 587/TCP, 25/TCP, 465/TCP (depending on the server)

Addresses list

- Licensing server (must be accessible when validating licenses of the IDE): optix-license-wapi.cloud.rockwellautomation.com
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